Generic Degree Two in $JC(2)$: Secant Projectors and Galois Rigidity

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Abstract

We study planar polynomial maps through the scheme of pairs with the same image. For

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}},$$

put $A = \mathbb{C}[x_1, x_2]$, let $I_R(F)$ be the collision ideal in the pair ring $S = \mathbb{C}[x_1, x_2, y_1, y_2]$, and let I_Δ be the diagonal ideal. Set $C_F = S/I_R(F)$. Their quotient

$$\mathrm{Obs}(F) = I_\Delta / I_R(F) \subseteq C_F$$

is the ideal of the diagonal in the collision ring. When F is Keller, an ordered divided-difference matrix produces an explicit idempotent $p_F \in C_F$ with $\mathrm{Obs}(F) = C_F p_F$. Using classical Galois rigidity, we characterize obstruction vanishing by normality of the induced function-field extension.

Writing $B = \mathbb{C}[P, Q] \subseteq A$, $K = \mathrm{Frac}(B)$, and $L = \mathrm{Frac}(A)$, suppose that the induced extension L/K has degree two. Then the generic collision algebra decomposes compatibly with the diagonal as

$$K \otimes_B C_F \simeq_K L \times L,$$

where diagonal evaluation is first projection. The second factor forces $\mathrm{Obs}(F) \neq 0$; when F is Keller, the secant identity $\mathrm{Obs}(F) = C_F p_F$ then gives $p_F \neq 0$. On the other hand, a separable quadratic extension is Galois, and classical Keller–Galois rigidity makes a hypothetical generic-degree-two Keller map a polynomial automorphism. Then $\mathrm{Obs}(F) = 0$ and $p_F = 0$, a contradiction. Thus no planar Keller map has generic degree two. The collision and secant arguments are formalized in Lean; the classical rigidity theorem is isolated there as the sole external mathematical input to the degree-two exclusion.

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1 Introduction

The complex two-dimensional Jacobian conjecture $JC(2)$ asks whether every polynomial map

$$F = (P, Q)\colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^{\times},$$

has a polynomial inverse. Such an F is called a *Keller map*. The condition on the Jacobian is local, while polynomial invertibility is global. Collision geometry makes the global issue visible by retaining pairs (x, y) for which $F(x) = F(y)$.

The companion paper [vdV26] develops the dimension-independent collision construction. It associates to a polynomial map two ideals

$$I_R(F) = (F_i(x) - F_i(y)), \qquad I_\Delta = (x_i - y_i)$$

inside the coordinate ring of ordered pairs. The containment $I_R(F) \subseteq I_\Delta$ induces the obstruction ideal

$$\mathrm{Obs}(F) = I_\Delta / I_R(F)$$

in the collision ring C_F. This ideal defines the diagonal inside the collision scheme. Its vanishing says that the collision scheme consists only of the diagonal; over $\mathbb{C}$, this is equivalent to polynomial automorphy.

The same construction has a generic field-theoretic form. Write

$$A = \mathbb{C}[x_1, x_2], \qquad B = \mathbb{C}[P, Q] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

For a Keller map, L/K is finite and separable, and

$$d_F = [L : K]$$

is the generic degree of F. At the generic point, Paper I gives the compatible isomorphism

$$K \otimes_B C_F \simeq_K L \otimes_K L,$$

with base-changed diagonal evaluation equal to tensor multiplication.

Paper I already uses classical Galois rigidity to exclude counterexamples of generic degrees one and two in every dimension [vdV26, Corollary 4.1]. The purpose here is to realize the quadratic exclusion explicitly in the planar collision ring. Moskowicz states a broader planar prime-degree exclusion [Mos24]; we do not use that argument. Our focus is the explicit collision certificate and its formal verification.

The planar projector construction has no generic-degree restriction. Its application to $d_F = 2$ uses two complementary descriptions of the collision obstruction. Ordered divided differences give the planar multivariate Bézoutian determinant and an idempotent p_F generating $\mathrm{Obs}(F)$. At the generic point, a separable quadratic extension gives

$$L \otimes_K L \simeq_L L \times L,$$

with tensor multiplication equal to the first-coordinate map. The second factor forces $\mathrm{Obs}(F) \neq 0$, and under the Keller condition this gives $p_F \neq 0$.

Classical Galois rigidity gives the opposite conclusion. A separable quadratic extension is normal, so a hypothetical planar Keller map with $d_F = 2$ would be a polynomial automorphism [Cam73, Raz79, Wri81]. Its obstruction and projector would vanish. The contradiction excludes generic degree two.

The marked normal closure gives a degree-independent vanishing criterion. Let N/K be the normal closure of L/K, with marked inclusion $\iota\colon L \hookrightarrow N$, and let H fix $\iota(L)$. For a planar Keller map, vanishing is equivalent to $N = \iota(L)$, or $H = 1$. This is the normal-branch counterpart to Paper I's nonnormal cubic construction.

Section 2 recalls, without reproving, the general construction needed here. Section 3 constructs the planar projector. Section 4 proves quadratic collision nonvanishing, and Section 5 combines it with Galois rigidity. Section 6 gives the marked-closure vanishing criterion. After the conclusion, Section 8 records the formalization.

The degree-two exclusion is not a proof of all of $JC(2)$. Remaining possible counterexamples have nonnormal function-field extensions of generic degree at least three. No normalization, boundary, or inertia construction enters the degree-two proof.

2 Inputs from Paper I

We use the dimension-independent construction of Paper I [vdV26]. To keep the two manuscripts notationally identical, let

$$F = (F_1, \dots, F_n)\colon \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}},$$

and put

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n],$$
$$I_R(F) = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n,$$
$$C_F = S/I_R(F), \qquad R_F = \operatorname{Spec} C_F.$$

Since $I_R(F) \subseteq I_\Delta$, diagonal substitution descends along the quotient map $q_F\colon S \twoheadrightarrow C_F$ to

$$\bar{\mu}_F\colon C_F \twoheadrightarrow A, \qquad [s] \longmapsto s(x, x).$$

The image of I_Δ is the ideal

$$\mathrm{Obs}(F) := q_F(I_\Delta) = I_\Delta / I_R(F) \subseteq C_F,$$

which defines the diagonal inside R_F.

The following two results recall Paper I's diagonal-kernel and automorphism criteria [vdV26, Proposition 2.2 and Theorem 2.7] and its generic collision bridge [vdV26, Theorem 2.5]. No proof of the general collision construction is repeated here.

Theorem 2.1 (Diagonal kernel and automorphism detection)**.** *For every complex polynomial map as above,*

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R(F).$$

Moreover,

$$F \text{ is a polynomial automorphism} \quad \Longleftrightarrow \quad I_R(F) = I_\Delta \quad \Longleftrightarrow \quad \mathrm{Obs}(F) = 0.$$

The implication $\mathrm{Obs}(F) = 0 \Rightarrow F$ is a polynomial automorphism uses Ax–Grothendieck [Ax69]. The degree-two exclusion needs only the elementary forward implication.

Let

$$B = \mathbb{C}[F_1, \ldots, F_n] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A),$$

and let

$$\mu_L \colon L \otimes_K L \longrightarrow L, \qquad \ell \otimes \ell' \longmapsto \ell\ell'$$

be tensor multiplication.

Theorem 2.2 (Generic collision bridge)**.** *If L/K is finite, then*

$$K \otimes_B A \simeq_K L$$

and there is a K-algebra isomorphism

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L$$

which carries the base-changed diagonal evaluation to μ_L. Furthermore, flatness of the localization $B \to K$ identifies

$$K \otimes_B \mathrm{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F)$$

as K-vector spaces.

These statements combine the fiber-product, generic-base-change, and diagonal-kernel results of Paper I. Their compatibility with μ_L marks which generic factor is the diagonal.

Return now to

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}.$$

For a Keller map, P, Q are algebraically independent and L/K is finite separable. We write

$$d_F := [L : K]$$

for its generic degree. The 2 in $JC(2)$ denotes ambient dimension; the separate condition $d_F = 2$ concerns the generic fiber. The nonvanishing argument below only uses a finite separable extension of degree two. The Keller condition enters through the nonzero constant Jacobian and, at the end, through Galois rigidity.

3 The explicit planar secant projector

The matrix below is the ordered multivariate Bézoutian matrix associated to F; compare [McK23, SS75]. We retain the term *secant matrix* because its determinant separates the diagonal collision sheet from its complement.

Let

$$F = (P, Q)\colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^{\times}.$$

Set

$$S = \mathbb{C}[x_1, x_2, y_1, y_2], \qquad \mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \qquad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

3.1 Ordered divided differences

For $H \in \mathbb{C}[x_1, x_2]$, define

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1},$$

$$D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients lie in S. The ordered telescoping identity is

$$H(x) - H(y) = D_1 H\,(x_1 - y_1) + D_2 H\,(x_2 - y_2).$$

Define

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \qquad M_F\big|_{\Delta} = JF.$$

Let

$$\delta_F := \det(M_F) \in S.$$

Proposition 3.1 (Secant annihilation)**.** *The secant determinant satisfies*

$$\delta_F I_{\Delta} \subseteq I_R(F), \qquad \delta_F\big|_{\Delta} = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by the adjugate matrix gives

$$\delta_F \mathbf{d} = \operatorname{adj}(M_F)\mathbf{r}.$$

Both entries on the right belong to $I_R(F)$. Since the entries of $\mathbf{d}$ generate I_{Δ}, the first assertion follows. Restriction to the diagonal gives

$$\delta_F\big|_{\Delta} = \det(M_F|_{\Delta}) = \det JF = c.$$

□

3.2 The idempotent

Let $\bar{\delta}_F := q_F(\delta_F)$ be the class of δ_F in C_F, and put

$$e_F := c^{-1}\bar{\delta}_F, \qquad p_F := 1 - e_F = 1 - c^{-1}\bar{\delta}_F.$$

The element e_F is the diagonal projector, while p_F is the off-diagonal collision projector.

Theorem 3.2 (Planar secant projector)**.** *With the notation above,*

$$e_F^2 = e_F, \qquad p_F^2 = p_F, \qquad e_F p_F = 0, \qquad e_F + p_F = 1.$$

Moreover,

$$\mathrm{Obs}(F) = C_F p_F, \qquad \mathrm{Ann}_{C_F}(\mathrm{Obs}(F)) = C_F e_F,$$

and p_F is the unique idempotent $u \in C_F$ satisfying $\mathrm{Obs}(F) = C_F u$. Hence

$$C_F \simeq A \times C_F p_F.$$

In particular,

$$p_F = 0 \quad \Longleftrightarrow \quad \mathrm{Obs}(F) = 0.$$

Proof. Proposition 3.1 gives

$$\bar{\delta}_F \,\mathrm{Obs}(F) = 0, \qquad \bar{\delta}_F - c \in \mathrm{Obs}(F).$$

Equivalently,

$$e_F \,\mathrm{Obs}(F) = 0, \qquad e_F - 1 \in \mathrm{Obs}(F).$$

Hence $e_F(e_F - 1) = 0$, so $e_F^2 = e_F$; the corresponding identities for $p_F = 1 - e_F$ follow.

The second relation says $p_F \in \mathrm{Obs}(F)$. If $j \in \mathrm{Obs}(F)$, then

$$j = (e_F + p_F)j = p_F j,$$

so $\mathrm{Obs}(F) = C_F p_F$. An element annihilates $C_F p_F$ exactly when its p_F-part is zero, giving

$$\mathrm{Ann}_{C_F}(\mathrm{Obs}(F)) = C_F e_F.$$

For uniqueness, let $u^2 = u$ and suppose $C_F u = C_F p_F$. Since $u \in C_F p_F$, one has $u p_F = u$; since $p_F \in C_F u$, one has $u p_F = p_F$. Thus $u = p_F$.

Complementary idempotents give

$$C_F \xrightarrow{\sim} C_F e_F \times C_F p_F, \qquad a \longmapsto (a e_F, a p_F).$$

Diagonal evaluation restricts to an isomorphism $C_F e_F \simeq A$: if $a = \bar{\mu}_F(r)$, then $a = \bar{\mu}_F(r e_F)$, while its kernel is $C_F e_F \cap \mathrm{Obs}(F) = 0$. Thus $C_F \simeq A \times C_F p_F$. Finally, $\mathrm{Obs}(F) = C_F p_F$ gives

$$p_F = 0 \Longleftrightarrow \mathrm{Obs}(F) = 0.$$

□

Abstractly, the existence of an idempotent splitting is expected: a Keller map is unramified, so its diagonal in the collision scheme is open, while a morphism between affine schemes has closed diagonal [The26b, The26a]. The theorem supplies the explicit Bézoutian representative and identifies the ideal it generates. The uniqueness just proved shows that the resulting p_F is independent of the ordering used to write the divided differences.

4 Quadratic generic collisions

We now work at the generic point. The essential field-theoretic fact is the degree-two marked-root decomposition.

4.1 The quadratic self-tensor product

Let L/K be finite separable of degree two. Paper I's marked-root tensor decomposition applies to a primitive generator α: it splits the marked root from the residual polynomial

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Here h_α has degree one, so its quotient algebra is another copy of L. Thus Paper I's general marked-root decomposition [vdV26, Theorem 2.6] specializes as follows.

Proposition 4.1 (Quadratic marked-root decomposition)**.** *There is an L-algebra isomorphism*

$$\Psi_\alpha \colon L \otimes_K L \xrightarrow{\sim} L \times L$$

under which tensor multiplication

$$\mu_L \colon L \otimes_K L \longrightarrow L$$

is first projection. Consequently,

$$\Psi_\alpha(\ker(\mu_L)) = 0 \times L \neq 0.$$

4.2 Nonvanishing of the affine obstruction

Combining the quoted generic collision bridge with Proposition 4.1 transports the nonzero second factor back to the affine obstruction.

Theorem 4.2 (Quadratic collision nonvanishing)**.** *Let $F \colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$ be a polynomial map whose induced extension L/K is finite separable of degree two. Then*

$$\mathrm{Obs}(F) \neq 0.$$

Equivalently,

$$I_R(F) \subsetneq I_\Delta.$$

Proof. Flatness of $B \to K$ gives a canonical identification

$$K \otimes_B \mathrm{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F).$$

Under the compatible quadratic product, the first-coordinate map has kernel $0 \times L$, which is nonzero. Therefore $K \otimes_B \mathrm{Obs}(F) \neq 0$, and hence $\mathrm{Obs}(F) \neq 0$. Since $\mathrm{Obs}(F) = I_\Delta/I_R(F)$, this is equivalent to strict containment. □

This theorem is independent of the Keller condition: generic degree two already forces a residual generic collision. For a planar Keller map, the secant projector makes that factor explicit.

Corollary 4.3 (Nonzero planar projector in generic degree two)**.** *Let*

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^{\times}.$$

If $d_F = 2$*, then*

$$p_F \neq 0.$$

Proof. The Keller condition makes L/K finite separable. Theorem 4.2 gives $\mathrm{Obs}(F) \neq 0$, and Theorem 3.2 gives $\mathrm{Obs}(F) = C_F p_F$. Therefore $p_F \neq 0$. □

5 Galois rigidity and the collision contradiction

We now use the classical Galois case of the Jacobian conjecture.

Theorem 5.1 (Classical Keller–Galois rigidity)**.** *Let*

$$F \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}$$

be a Keller map. If the induced finite separable extension L/K is normal, equivalently Galois, then F is a polynomial automorphism.

This is the classical normal/Galois case [Cam73, Raz79, Wri81]; we use it as a literature theorem and do not reprove it here.

Every separable extension of degree two is normal: if the minimal polynomial of a primitive generator has one root in L, its other root also lies in L. Thus Theorem 5.1 applies automatically on the generic-degree-two Keller locus.

Proposition 5.2 (Rigidity forces collision vanishing)**.** *Let*

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF = c \in \mathbb{C}^{\times}.$$

If $d_F = 2$*, then Keller–Galois rigidity gives*

$$\mathrm{Obs}(F) = 0, \qquad p_F = 0.$$

Proof. The extension L/K is finite separable of degree two, hence Galois. Theorem 5.1 makes F a polynomial automorphism. Theorem 2.1 then gives

$$I_R(F) = I_{\Delta}, \qquad \mathrm{Obs}(F) = 0.$$

Finally, Theorem 3.2 gives $p_F = 0$. □

Theorem 5.3 (Exclusion of generic degree two)**.** *No planar Keller map has generic degree two. Equivalently, every polynomial map*

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^{\times},$$

satisfies

$$d_F \neq 2.$$

Proof. Suppose that $d_F = 2$. Corollary 4.3, which reads the residual factor in

$$K \otimes_B C_F \simeq_K L \times L,$$

gives $p_F \neq 0$. Proposition 5.2 gives $p_F = 0$. This is a contradiction. □

Equivalently, the two halves of the proof say

$$d_F = 2 \Longrightarrow \mathrm{Obs}(F) \neq 0$$

by the quadratic generic collision product, while

$$F \text{ Keller and } L/K \text{ normal} \Longrightarrow \mathrm{Obs}(F) = 0$$

by Galois rigidity and automorphism detection. The secant construction expresses both statements as nonvanishing and vanishing of one canonical idempotent.

6 Vanishing and the marked normal closure

The same marked normal closure as in Paper I characterizes when the planar projector vanishes. Throughout this section, $F = (P, Q)$ is a planar Keller map, with no generic-degree restriction.

Let N be the normal closure of L/K, fix the embedding $\iota\colon L \hookrightarrow N$, and put

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/\iota(L)).$$

These are the marked objects of Paper I.

Proposition 6.1 (Normal-closure criterion)**.** *For a planar Keller map, the following conditions are equivalent:*

$$p_F = 0 \Longleftrightarrow \mathrm{Obs}(F) = 0 \Longleftrightarrow L/K \text{ is normal}$$
$$\Longleftrightarrow N = \iota(L) \Longleftrightarrow H = 1.$$

Equivalently, every conjugate source coordinate satisfies $g(\iota(x_i)) \in \iota(L)$ for $g \in G$ and $i = 1, 2$.

Proof. Theorem 3.2 identifies $p_F = 0$ with $\mathrm{Obs}(F) = 0$. If $\mathrm{Obs}(F) = 0$, then $I_R(F) = I_\Delta$. Evaluating at the collision pair $(\iota(x), g(\iota(x)))$ gives $g(\iota(x_i)) = \iota(x_i)$ for every $g \in G$ and $i = 1, 2$. The conjugate source coordinates generate N over K, so $N = \iota(L)$. This equality is equivalent to normality of L/K and, by Galois correspondence, to $H = 1$. Conversely, normality gives automorphy by Theorem 5.1, hence $\mathrm{Obs}(F) = 0$. The same generation statement shows that containment of all conjugate source coordinates in $\iota(L)$ suffices. □

Normality of N/K is supplied by the construction; normality of L/K is the criterion. In the cubic nonnormal case of Paper I, $\iota(L) \subsetneq N$ and $H \neq 1$. Here the complementary condition $N = \iota(L)$, together with Keller rigidity, forces $L = K$. In generic degree two, normality is automatic, so this criterion gives the vanishing half of the contradiction in Theorem 5.3. It does not assert vanishing for every planar Keller map.

7 Conclusion

For a planar Keller map, the divided-difference determinant gives

$$C_F \simeq A \times C_F p_F, \qquad \mathrm{Obs}(F) = C_F p_F,$$

where p_F is the explicit off-diagonal collision projector. Its vanishing has the marked-closure description

$$p_F = 0 \iff \mathrm{Obs}(F) = 0 \iff N = \iota(L) \iff H = 1.$$

This is the normal-branch counterpart to the nonnormal cubic construction of Paper I.

In generic degree two, the residual factor forces $p_F \neq 0$, while normality and Keller–Galois rigidity force $p_F = 0$. The incompatibility excludes that degree. Remaining possible counterexamples to $JC(2)$ have nonnormal function-field extensions of generic degree at least three. Uniform planar vanishing beyond the quadratic case is a separate research problem; no boundary or ramification criterion is needed for the exclusion proved here.

8 Lean formalization status

Lean correspondence. The main development uses Lean 4.28.0 and Mathlib v4.28.0 [dMU21, The20]. The checked submission release uses Lean 4.35.0 and Mathlib v4.35.0. Shared collision objects and generic base change remain in `CollisionIdeals/General/Collision` and `CollisionIdeals/General/GenericFiber`; the planar constructions are in `CollisionIdeals/Planar/ExplicitSecant.lean` and `CollisionIdeals/Planar/GenericDegreeTwo/Main.lean`.
The compared declarations are `CollisionIdeals.Palomar.PaperTwo.explicitPlanarSecantProjector` for the idempotence, obstruction-generator, annihilator, and uniqueness claims of Theorem 3.2, and `CollisionIdeals.Palomar.PaperTwo.quadraticPlanarCollisionRigidity` for Theorem 5.3. The product splitting in Theorem 3.2 is the standard consequence of the complementary idempotents, not a separately compared declaration.
At release commit `ccbba13`, the protected preflight accepted both declarations with Comparator, Lean, NanoDa, and con-ron. The reported axioms are `propext`, `Classical.choice`, and `Quot.sound`. This verifies the selected formal statements, not a proof of the full planar conjecture.
The degree-two exclusion takes `ComplexKellerGaloisRigidity 2` explicitly as its sole external mathematical input. It represents Theorem 5.1, rather than an installed axiom or a theorem proved here. The degree-two vanishing step is formalized in `CollisionIdeals/Planar/GenericDegreeTwo/Vanishing.lean`.
The components of Proposition 6.1 are proved in the main development in `CollisionIdeals/Planar/Research/Vanishing/GaloisDescent.lean` and the shared normal-closure API; they are separate from the two compared declarations. Vanishing implies normality directly through collision equations, using `CollisionIdeals/General/Galois/CollisionVanishing.lean`; only the converse uses Galois rigidity. The general automorphism equivalence quoted from Paper I retains its Ax–Grothendieck input.

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